

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Seat No.

Midterm Examination of Semester 1

Year: 2019

Subject: 392133 Physics III

Section: 15-18

Date: 16 August 2019

Time: 13.00-15.00

Name: _____ ID: _____ Class: _____

Directions: The test is designed to measure your comprehension. The test is divided into 1 sections. There will be 7 pages (including this page) and they are worth 60 points (12 problems).

- Answer the questions on this test papers.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators cannot be used in this test.

Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

1. [5 points] A block attached to a spring oscillates in simple harmonic motion along the x axis. The limits of its motion are $x = 10$ cm and $x = 60$ cm and it goes from one of these extremes to the other in 0.25 s. Find its amplitude and frequency.

วิทยาลัยเทคโนโลยีอุตสาหกรรม

2. [5 points] A weight from an ideal spring oscillates up and down with a period T . If the amplitude of the oscillation is now doubled, find the period in the term of T .

3. [5 points] A particle moves in simple harmonic motion according to $x = 2\cos(50t)$, where x is in meters and t is in seconds. Determine its maximum velocity in m/s.

วิทยาลัยเทคโนโลยีอุตสาหกรรม

4. [5 points] A 0.20 kg object mass attached to a spring whose spring constant is 500 N/m executes simple harmonic motion. If its maximum speed is 5.0 m/s, determine the amplitude of its oscillation.

5. [5 points] Water waves in the sea are observed to have a wavelength of 300 m and a frequency of 0.07 Hz. Find the speed of these waves.

6. [5 points] The tension in a string with a linear density of 0.0010 kg/m is 0.40 N. If a 100 Hz sinusoidal wave on this string, find its wavelength.

วิทยาลัยเทคโนโลยี
และอาชีวศึกษา
ชลบุรี

7. [5 points] A piano wire has length L and mass M . If its fundamental frequency is f , show that its tension F_T is:

$$F_T = 4LMf^2$$

8. [5 points] Two small identical speakers are connected (in phase) to the same source. The speakers are 3 m apart and at ear level. An observer stands at X, 4 m in front of one speaker as shown. If the amplitudes are not changed, find the wavelength of the sound that he hears at the most intense.



9. **[5 points]** The intensity of sound wave A is 100 times that of sound wave B. If the sound level of wave B is 10 dB, determine the sound level of wave A.

10. **[5 points]** When listening to tuning forks of frequency 256 Hz and 260 Hz, find the number of beats per second.

11. [5 points] A vibrating tuning fork is held over a water column with one end closed and the other open. As the water level is allowed to fall, a loud sound is heard for water levels separated by 17 cm. If the speed of sound in air is 340 m/s, determine the frequency of the tuning fork.

12. [5 points] A stationary source emits a sound wave of frequency f . If it were possible for a man to travel toward the source at the speed of sound, determine the frequency of the emitted sound he would observe.

วิทยาลัยเทคโนโลยีอุตสาหกรรม

Asst.Prof.Dr.Amarin Ratanavis

Dr. Thanakorn Iamsasri